Docker部署ELK

ELK是指Elasticsearch、Kibana、Logstash这三种服务搭建的日志收集系统，点击进入官方下载网站。

ELK中各个服务的作用：

- Elasticsearch：用于存储收集到的日志信息；
- Logstash：用于收集日志，SpringBoot应用整合了Logstash以后会把日志发送给Logstash,Logstash再把日志转发给Elasticsearch；
- Kibana：通过Web端的可视化界面来查看日志。

安装参考：

- https://www.elastic.co/guide/en/elasticsearch/reference/8.19/docker.html
- https://www.elastic.co/guide/en/kibana/8.19/docker.html
- https://www.elastic.co/guide/en/logstash/8.19/docker.html

在生产环境中在Docker中运行Elasticsearch时适用以下要求和建议，参考下面链接：

https://www.elastic.co/guide/en/elasticsearch/reference/8.19/docker.html#docker-prod-prerequisites

提示：本教程以8.19.14版本为例，如果要安装其他版本，切换镜像版本号即可

1 准备目录

在home目录下面创建`elk`目录

```
mkdir -p /home/elk/8.19.14
cd /home/elk/8.19.14
mkdir logstash elasticsearch elasticsearch/data elasticsearch/plugins
# 修改数据目录访问权限
chown 1000:1000 elasticsearch/data elasticsearch/plugins
```

将资源目录下面的`logstash.conf`上传到`logstash`目录下面。

2 开放端口

```
# 开放端口
firewall-cmd \
--add-port 9200/tcp \
--add-port 9300/tcp \
--add-port=4560-4563/tcp \
--add-port=5601/tcp \
--permanent
# 重新加载防火墙
firewall-cmd --reload
```

3 服务编排

执行命令行`vi docker-compose.yml`创建服务编排文件，然后在文件中写入下列内容后保存。

当然你也可以使用资源目录中提供`yml`文件直接上传到`elk`目录即可，然后跳过本步骤。

```yaml
services:
  elasticsearch:
    image: docker.elastic.co/elasticsearch/elasticsearch:8.19.14
    container_name: elasticsearch
    environment:
      - "node.name=elasticsearch"
      - "cluster.name=elk-cluster"
      - "discovery.type=single-node"
      - "xpack.security.enabled=false"
      - "ES_JAVA_OPTS=-Xms256m -Xmx256m"
      - TZ=Asia/Shanghai
    volumes:
      - ./elasticsearch/plugins:/usr/share/elasticsearch/plugins
      - ./elasticsearch/data:/usr/share/elasticsearch/data
    ports:
      - 9200:9200
      - 9300:9300
    networks:
      - elk
  kibana:
    image: docker.elastic.co/kibana/kibana:8.19.14
    container_name: kibana
    depends_on:
      - elasticsearch
    environment:
      - ELASTICSEARCH_HOSTS=http://elasticsearch:9200
      - I18N_LOCALE=zh-CN
      - "JAVA_OPTS=-Xms256m -Xmx256m"
      - TZ=Asia/Shanghai
    ports:
      - 5601:5601
    networks:
      - elk
  logstash:
    image: docker.elastic.co/logstash/logstash:8.19.14
    container_name: logstash
    environment:
      - "LS_JAVA_OPTS=-Xms256m -Xmx256m"
      - TZ=Asia/Shanghai
    volumes:
      - ./logstash/logstash.conf:/usr/share/logstash/pipeline/logstash.conf:ro
    depends_on:
      - elasticsearch
    ports:
      - "4560-4563:4560-4563"
    networks:
      - elk
networks:
  elk:
    name: elk
    driver: bridge
```

4 启动服务

执行命令`docker compose up`。

首次启动会拉取镜像，耐心等待镜像下载完成。

如果启动过程没有看到明显的错误，那么表示服务器启动成功，接下来访问一下elasticsearch和kibana。

首先访问elasticsearch，访问地址：http://ip:9200 如果看了类似下图所示的结果表示elasticsearch启动成功。



接下来在访问kibana，访问地址：http://ip:5601 如果看了类似下图所示的结果表示kibana启动成功。



界面上点 **自己浏览** 可跳过繁琐的集成引导。

然后终端执行`Ctrl + C`结束前台启动，成功停止效果如下图所示

```
^CGracefully stopping... (press Ctrl+C again to force)
[+] Stopping 3/3
 ✔ Container logstash        Stopped
 ✔ Container kibana          Stopped
 ✔ Container elasticsearch   Stopped
canceled
```

执行后台启动，使用命令`docker compose up -d`，启动成功如下图所示

```
[root@7c10c9d2eb8d 8.19.14]# docker compose up -d
[+] Running 4/4
 ✔ Network 81914_default     Created
 ✔ Container elasticsearch   Started
 ✔ Container kibana          Started
 ✔ Container logstash        Started
```

使用`docker ps`指令查看进程

```
[root@7c10c9d2eb8d 8.19.14]# docker ps
CONTAINER ID   IMAGE                                                  COMMAND                  CREATED              STATUS
5e6962c75a67   docker.elastic.co/logstash/logstash:8.19.14            "/usr/local/bin/dock…"   About a minute ago   Up About a minute
66b1a7bc87c9   docker.elastic.co/kibana/kibana:8.19.14                "/bin/tini -- /usr/l…"   About a minute ago   Up About a minute
e96dde8dee82   docker.elastic.co/elasticsearch/elasticsearch:8.19.14  "/bin/tini -- /usr/l…"   About a minute ago   Up About a minute
```

5 安装插件

5.1 安装IK分词插件

进入elasticsearch容器

```
docker exec -it elasticsearch /bin/bash
```

执行插件安装命令

```
elasticsearch-plugin install https://release.infinilabs.com/analysis-ik/stable/elasticsearch-analysis-ik-8.19.14.zip
```

安装完成效果如下图所示

```
elasticsearch@dbf382499c57:~$ elasticsearch-plugin install https://release.infinilabs.com/analysis-ik/s
-> Installing https://release.infinilabs.com/analysis-ik/stable/elasticsearch-analysis-ik-8.19.14.zip
-> Downloading https://release.infinilabs.com/analysis-ik/stable/elasticsearch-analysis-ik-8.19.14.zip
[=================================================] 100%??
WARNING: this plugin contains a legacy Security Policy file. Starting with version 8.18, Entitlements r
ps://www.elastic.co/guide/en/elasticsearch/plugins/current/creating-classic-plugins.html
@@@
```

5.2 重启服务

所有插件安装完成统一重启一下服务。

```
elasticsearch@a385d58bbca0:~$ exit
exit
[root@7c10c9d2eb8d 8.19.14]# docker compose restart
[+] Restarting 3/3
 ✔ Container kibana         Started
 ✔ Container elasticsearch  Started
 ✔ Container logstash       Started
[root@7c10c9d2eb8d 8.19.14]#
```